

TAICHUNG, Taiwan

WMO: 467490

Lat: 24.1457N

Lon: 120.6840E

Elev: 276

StdP: 14.55

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	49.8	52.3	38.4	34.6	54.8	41.8	39.4	56.0	12.4	62.3	11.0	63.1	4.5	10	0.280

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	92.5	79.8	91.5	79.4	90.5	79.0	81.4	89.7	80.7	89.1	80.0	88.2	5.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.1	152.7	86.2	78.3	148.3	85.6	77.7	145.2	85.2	45.5	90.0	44.6	89.3	43.9	88.6	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.6	9.5	8.4	DB	46.4	95.4	3.1	2.2	44.2	97.0	42.4	98.3	40.7	99.6	38.5	101.2	
			WB	42.7	82.8	2.8	0.8	40.6	83.4	39.0	83.9	37.4	84.3	35.3	84.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.5	63.6	64.8	68.8	75.1	79.8	82.9	84.4	83.6	82.6	79.0	73.6	66.6	(6)
	DBStd	8.49	4.95	5.86	5.68	4.98	3.50	2.97	2.34	2.36	2.57	3.22	4.11	4.92	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	229	83	70	29	4	0	0	0	0	0	0	2	41	(9)
	CDD50	9293	423	416	583	752	923	987	1066	1041	979	900	708	515	(10)
	CDD65	4046	41	65	147	306	458	537	601	576	529	435	260	91	(11)
	CDH74	38472	104	220	665	2032	4256	6103	7384	6746	5847	3581	1287	247	(12)
	CDH80	13790	5	22	102	458	1332	2423	3283	2733	2176	1013	224	19	(13)
(14) Wind	WSAvg	4.1	4.3	4.2	4.1	3.8	3.8	4.1	4.2	3.7	3.9	4.3	4.0	4.3	(14)
(15) Precipitation	PrecAvg	63.1	1.3	3.0	3.7	4.6	8.4	13.7	9.2	12.4	4.3	0.6	0.7	0.8	(15)
	PrecMax	92.0	4.8	15.1	13.9	18.6	21.3	35.0	25.3	30.9	22.7	5.1	2.9	3.3	(16)
	PrecMin	38.5	0.2	0.1	0.1	0.0	0.2	1.8	0.8	2.9	0.2	0.0	0.0	0.0	(17)
	PrecStd	13.9	1.1	3.7	2.8	3.9	4.7	8.6	5.9	6.7	4.3	0.9	0.8	0.9	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	80.2	82.9	85.7	88.7	91.7	93.3	94.0	93.3	92.5	90.4	87.0	82.6	(19)
		MCWB	67.3	70.3	73.0	76.1	78.6	80.0	80.1	80.4	79.4	78.1	74.5	69.5	(20)
	2%	DB	76.9	79.5	82.7	86.6	89.9	91.8	92.5	91.7	90.8	88.4	84.6	79.2	(21)
		MCWB	65.5	68.3	71.4	74.9	77.7	79.5	79.8	79.8	78.9	77.0	73.2	68.0	(22)
	5%	DB	74.2	76.8	80.4	84.6	88.4	90.7	91.4	90.6	89.7	86.8	82.7	76.8	(23)
		MCWB	64.3	66.4	70.0	74.0	77.1	79.1	79.5	79.4	78.4	75.9	72.1	66.5	(24)
	10%	DB	71.3	73.9	77.8	82.8	86.6	89.3	90.3	89.4	88.4	85.3	80.6	74.4	(25)
		MCWB	62.8	65.3	68.6	73.0	76.2	78.6	79.1	79.0	77.8	74.8	70.6	65.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.7	71.9	74.6	77.7	80.9	81.9	82.5	82.5	81.3	79.9	76.5	71.9	(27)
		MCDB	76.5	80.1	83.3	86.3	89.3	91.2	90.5	90.0	89.8	88.6	83.9	79.7	(28)
	2%	WB	67.3	69.8	72.8	76.1	79.2	80.7	81.2	81.3	80.2	78.4	74.9	69.9	(29)
		MCDB	74.3	76.9	80.2	84.0	87.4	89.9	89.7	89.0	88.3	86.1	81.9	76.2	(30)
	5%	WB	65.6	67.9	71.4	75.0	78.1	79.8	80.4	80.4	79.5	77.3	73.7	68.2	(31)
		MCDB	72.0	74.7	78.3	82.6	85.9	88.5	89.0	88.1	87.3	84.7	80.4	74.6	(32)
	10%	WB	64.0	65.9	69.9	74.0	77.2	79.0	79.7	79.7	78.8	76.3	72.4	66.7	(33)
		MCDB	69.9	73.0	76.7	81.2	84.5	87.3	88.1	87.2	86.3	83.4	78.9	72.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	12.8	13.1	12.0	11.2	10.9	11.6	10.8	11.3	12.2	12.9	13.3	(35)
		MCDBR	17.0	17.3	16.1	13.9	13.2	13.0	12.8	12.5	12.6	13.1	14.7	16.2	(36)
		MCWBR	8.0	8.3	7.6	6.1	5.7	5.8	5.2	5.1	4.9	5.7	6.6	7.8	(37)
	5% WB	MCDBR	14.9	15.8	14.3	12.6	11.6	11.9	12.1	11.6	11.5	11.4	12.8	13.5	(38)
		MCWBR	7.7	8.4	7.3	6.1	5.6	5.8	5.4	5.3	4.8	5.2	6.1	7.0	(39)
(40) Clear-Sky Solar Irradiance	taub		0.492	0.547	0.669	0.681	0.584	0.536	0.549	0.585	0.579	0.541	0.503	0.494	(40)
	taud		2.010	1.895	1.649	1.678	1.936	2.084	2.037	1.929	1.923	1.971	2.061	2.023	(41)
	Ebn at Noon		239	234	213	215	236	246	242	234	232	233	234	232	(42)
	Edh at Noon		50	60	80	79	61	52	55	61	60	55	47	48	(43)
(44) All-Sky Solar Radiation	RadAvg		922	1023	1171	1284	1469	1620	1843	1668	1536	1359	1025	911	(44)
	RadStd		124	147	132	174	220	177	116	163	145	113	96	110	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.45	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	+229	+202

Nomenclature: See separate page